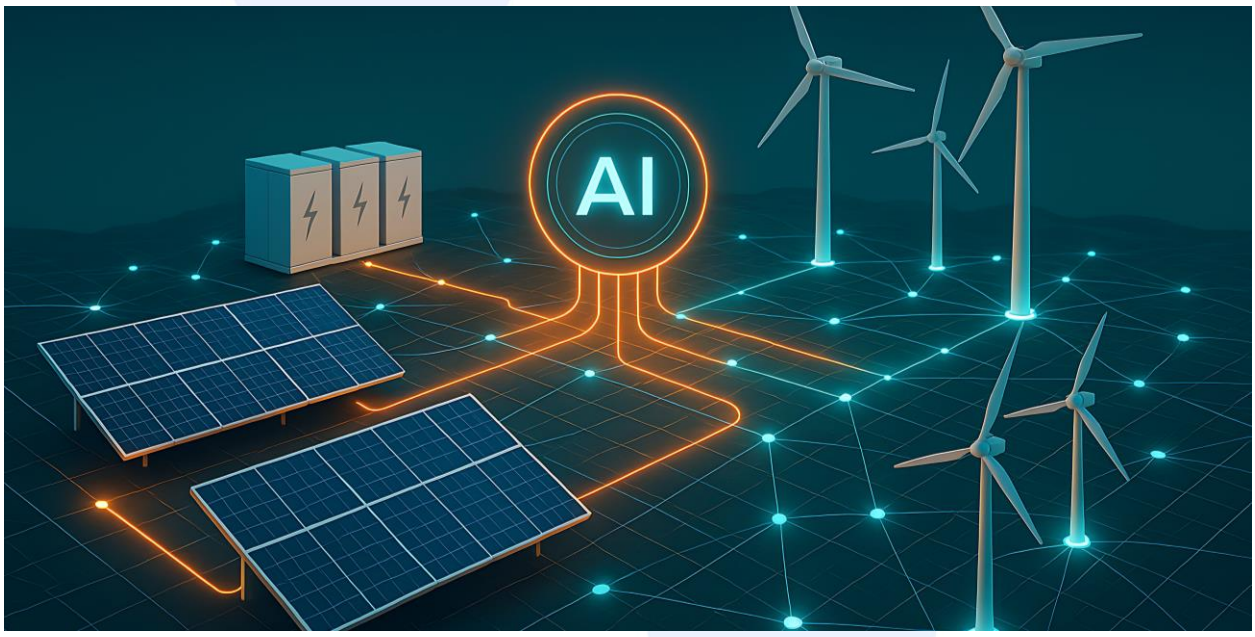
European Commission

Joint Research Centre

How AI is shaping EU electricity grids: the impact of AI Act

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Contents

Abstract 3

Acknowledgements 4

Authors 4

Executive summary 5

Policy context 5

Key conclusions 5

Main findings 6

Related and future Joint Research Centre work 6

Quick guide 7

1. Introduction 8

1.1. Policy background 8

1.2. AI Act 9

1.3. AI Act on the energy field 10

2. Methodology 11

2.1. Introduction 11

2.2. Awareness of AI Act 11

2.3. Use of the AI in the energy sector 12

2.4. Energy efficiency and environmental impact 12

2.5. Best practices, documentation and ethics 13

2.6. Codes of Conduct 14

2.7. Social implications 14

2.8. Other remarks and future involvement 15

3. Survey analysis and results 16

3.1. Introduction 16

3.2. Awareness and adaptation to the AI Act 17

3.3. Use of AI 18

3.4. Energy efficiency and environmental impact 19

3.5. Best practices 21

3.6. Codes of conduct 22

4. Conclusions 24

References 25
List of abbreviations and definitions 27
List of figures 28

Abstract

This report gives a first overview of the implementation of the Artificial Intelligence (AI) Act by the European Distribution System Operators (DSOs). Given the increase in the uptake of AI tools by the energy sector and the fact that energy is the field that can offer a variety of use cases, where AI tools can be applied, it is fundamental to study the effect of the recently published AI Act. This report collects actual data from European DSOs aiming at depicting the situation with respect to the AI Act Implementation and at identifying gaps in the field.

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Authors

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Executive summary

In this report, data regarding the implementation of the AI Act by some of the European Distribution System Operators (DSOs) were collected and analysed. Insights from DSOs concerning the implementation of the Artificial Intelligence Act in the energy sector were gathered through a dedicated EU survey. The feedback collected focuses on the current uses of AI, compliance strategies, environmental and societal impacts, and stakeholder engagement. The results are expected to support further analyses on how AI is transforming energy operations and where additional guidance, or collaboration may be needed.

Policy context

This report is directly linked to the Artificial Intelligence Act, aiming to set the legal rules for artificial intelligence technologies in Europe.

The AI Act itself is linked to specific policies of the European Commission, among others:

- European Digital Strategy: being part of this strategy, the AI Act aims to make Europe a leader in digital innovation.
- General Data Protection Regulation (GDPR): AI Act complements this regulation by ensuring that AI tools are compliant with data protection regulations.
- European Green Deal: the AI Act promotes the responsible use of AI to support the EU's climate and sustainability objectives, including the energy transition and the move towards climate neutrality.

This report examines the implementation of the AI Act in the energy sector, by European DSOs, thus aligning with Draghi report [1], which states that the energy sector is a field that may offer a vast variety of use cases, where AI tools may be used.

Key conclusions

The implementation of the AI Act provides a crucial framework for the regulation of AI technology in the EU, with significant implications for the energy sector. Throughout the report, several policy-relevant conclusions emerged:

- Policy implementation: the AI Act's assumptions about AI safety and ethical utilization have been confirmed by the DSOs analysed in this study, as they recognize the effectiveness in addressing risks, promoting transparency, and fostering the responsible use of AI technologies. However, ongoing challenges in terms of interpretation and practical compliance are still faced by most of the DSOs analysed.
- Emerging problems and adjustments: confusion over specific terminologies like "deployer" and "provider" and ambiguity in certification procedures highlight areas requiring clarification. Policymakers should consider adjusting definitions within the Act to enhance its clarity.

- Social and environmental impacts: AI's application in the energy sector is revealing significant positive impacts, such as operational efficiency and enhanced sustainability efforts. However, the increased energy consumption by AI models and the location of data centres closer to urban areas necessitate a reassessment of costs and benefits considering both societal and environmental implications.
- Knowledge gaps and uncertainties: the report identifies uncertainties regarding AI's deployment for critical infrastructure and risk management in customer interactions.
- Potential for innovation: there is considerable potential for AI-driven innovations in energy efficiency and renewable energy sources (RES) integration. However, these must be undertaken with caution to mitigate risks associated with data security and ethical concerns.

Main findings

- Awareness and compliance with AI Act: 90% of DSOs are aware of the AI Act, reflecting a significant commitment to regulatory compliance, but the whole understanding does not match with the effective operational application, indicating a need for further support to ensure practical compliance.
- Application in energy sector: AI is predominantly used for demand forecasting and grid management, with 60% of DSOs employing AI in these areas. Its adoption is enhancing operational efficiency, although there are active efforts to address considerations around energy consumption and optimizing human-AI interaction.
- Energy efficiency and environmental impact: AI applications are contributing to sustainability goals, with 70% of DSOs leveraging AI for energy optimization. However, while AI can help in optimizing energy use, the reductions of direct emission using AI-driven technologies is still a developing area. DSOs are exploring new methodologies and technologies to enhance AI's capacity for effective emission monitoring and management.
- Best practices and documentation: documentation practices vary, with some DSOs maintaining comprehensive records and others in early organizational stages. Alignment with best practices is a proactive step towards fostering responsible AI deployment.
- Codes of conduct and pilot testing: AI codes of conduct are limited, but many DSOs have been engaged in real-world testing.

Related and future Joint Research Centre work

The JRC is active in the field of Artificial Intelligence and its role in the energy sector. An ongoing work is the Code of Conduct for Data Centres (EU DC CoC), which is a voluntary initiative to guide data centre operators on reducing energy consumption without compromising the functionality of such facilities [2]. Another ongoing work of the JRC is a literature review about the impact of AI on the future energy system, covering among others, increased energy consumption and possible greenhouse emission savings. Furthermore, JRC is active in investigating social implications of AI use in energy sector through an energy justice lens [3].

In addition, the JRC holds several activities in the field of artificial intelligence, mainly through its AIDA portfolio [4], contributing to AI and data science research, supporting AI and data policies and

fostering the development of research infrastructures and platforms to enhance the effective use of data and AI.

Quick guide

Chapter 1 introduces the policy context, presents the AI Act and its implications for the energy field. Chapter 2 gives the methodology used to collect the data used, while Chapter 3 provides the results of our survey on the implementation of AI by the European DSOs. Chapter 4 concludes the report.

1. Introduction

1.1. Policy background

The initial proposal for the Artificial Intelligence Act was introduced in April 2021 by the European Commission. The scope has been to regulate AI technologies within the European Union, thus guaranteeing their safety and ethical perspective. The AI Act (Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence) [5] is the very first legal framework on AI at global level. It addresses the risks of AI and puts Europe in a leading role to this respect. Its overall scope has been to foster the usage of AI tools, whereas maintaining safety and data protection and avoiding misuse or possible abuse of AI tools, which would undermine the initial planned benefits.

Before the establishment of the AI Act, the EU created ethical guidelines for trustworthy AI, which were used to pave the way for the AI Act. These guidelines highlight the importance of fundamental human rights, like privacy and non-discrimination. The AI Act fosters product safety and liability, by introducing requirements to ensure the safety of AI systems and by complementing directives like the Product Liability Directive.

The AI Act itself is linked to and supports several policies of the European Commission:

- European Digital Strategy [6]:

First of all, the AI Act is part of the European Digital Strategy, aiming to make Europe a leader in digital innovation. The Digital Strategy promotes the use of advanced technologies, like AI and cloud computing in several sectors, while promoting sustainability. It is an initiative that fosters digital technologies in order to enhance EU's economic resilience and digital sovereignty. It promotes the usage of modern infrastructure and tools, including AI tools, whereas promoting collaboration and skills development to reach the ambitious goals set for 2030. Therefore, the AI Act complements the European Digital Strategy, as it fosters the correct usage of AI tools ensuring data protection and safety.

- General Data Protection Regulation (GDPR) [7]:

The GDPR regulation is related to the protection of personal data and privacy. The AI Act complements this regulation by ensuring that AI tools are compliant with data protection regulations. In the era of digital transition and with the growth of AI tools, it is fundamental to set rules for personal data protection.

- European Green Deal [8]:

The European Green Deal is an umbrella policy, which aims at promoting environmental goals and has ambitious targets for 2030 and 2050. Digital technologies, including AI tools, play a key role in many use cases in various sectors, facilitating data processing, patterns extraction and forecasts. Therefore, AI tools play a crucial role in the effort to reduce the CO_2 footprint and reach the overall climate and sustainability goals. Their correct functionality and usage are fostered by the AI Act, which helps use case developers to use AI tools responsibly.

- Digital Services Act (DSA) [9], and Digital Markets Act (DMA) [10]:

The AI Act complements these two acts, which focus on the online platforms and digital markets. Indeed, the AI Act ensures that the AI systems used are safe, thus contributing to the fair functionality of these Acts.

1.2. AI Act

In this section, an overview of the AI Act and its main points is provided. To begin with, it has been designed and developed to ensure the safe usage of AI systems while respecting existing regulations on fundamental rights and values.

The AI systems are divided into four categories depending on the risk level:

- Unacceptable risk: such systems are classified as dangerous for the safety and for not respecting human rights. They are not allowed under the AI Act. Examples are the systems that use techniques to take advantage of vulnerabilities of specific groups or systems that distort people's behaviour.
- High-risk AI systems: these systems are related to AI applications used for example in critical infrastructure, education, employment and law enforcement. Such AI systems need to fulfil specific requirements before they can be employed, and their developers should register these systems in the EU database. Specific standards for data quality, transparency, accuracy and robustness need to be met.
- Limited risk AI systems: the systems under this category do not entail significant risk, but they need to fulfil specific obligations, i.e. in case there is interaction with people, they should be aware that they are interacting with an AI system.
- Minimal risk AI systems: most AI systems fall under this category, entailing minimal risk and having the necessity to comply to minimum regulation.

Main points of the AI Act:

- Transparency: the AI Act highlights the necessity for awareness raising on behalf of users. Humans that interact with AI systems need to know how they function, especially for high-risk AI systems to minimize risks.
- Data Governance: the AI Act indicates that data needs to be processed with accurate methods ensuring that data sets are error-free and without bias.
- Penalties and enforcement: the AI Act imposes penalties for those who infringe the rules set for AI systems, with fines going up to €30 million or 6% of global annual turnover.
- Regulatory sandboxes: innovation is encouraged, especially through provisions for regulatory sandboxes, where innovative AI solutions can be tested with fewer regulatory constraints.
- International collaboration: the AI Act supports international collaboration and giving rules for any provider that operates its system on the EU market, even if it is located outside the EU.

- Adaptability and technology agnostic: the AI Act is technology neutral and it can be adaptable to new AI technologies, whereas it has the margin to evolve as AI techniques become more modern.

1.3. AI Act on the energy field

AI technology can be proved to be a powerful tool for the energy sector, as it is one of the fields that offers a vast variety of use cases. In the Draghi report [1] it is stated that: "*AI solutions already provide more than 50 use cases in energy systems today*" and "*with estimates of the market value for AI applications in the energy sector ranging up to USD 13 billion, the energy sector is one of the sectors with the greatest potential to benefit from the capacity of AI to boost energy efficiency and accelerate innovation*".

Similarly, the International Energy Agency (IEA) highlights in its Energy and AI report [11] that AI can play a key role in improving energy efficiency, supporting the integration of renewables, and enhancing grid reliability.

AI solutions are therefore fundamental for the energy sector, facilitating various applications, like:

- Demand side flexibility services,
- Predictive algorithms that can be used to forecast energy generation and demand,
- Smart charging of electric vehicles,
- Energy smart appliances for energy transition and consumer participation, including multiple categories of energy smart appliances

The AI Act is therefore crucial for the correct usage and functionality of AI tools, which has been one of the main drivers for this report. The main scope of this report is to depict the practical usage of the AI Act in the energy sector, and in particular by some of the European DSOs.

The objective is to see the extent to which the AI Act is being used, identify challenges for its implementation, limiting factors and cases of possible misinterpretation by the use case developers.

2. Methodology

2.1. Introduction

The recent introduction of the AI Act in the EU requires examining its practical implementation in the energy sector. European DSOs, who deliver energy to consumers, offer clear examples of how the regulation is applied.

As previously discussed, the energy sector offers numerous AI use cases, making it a critical area for application. The AI Act categorizes AI systems and sets guidelines for developers to ensure safety and data protection while preventing misuse. This section details the approach for evaluating DSOs application of the AI Act, assessing its usage and impact. A questionnaire was developed and distributed to European DSOs to track AI Act implementation and identify challenges and gaps. The survey is organized into sections addressing various issues related to the AI Act, with a brief description provided below.

2.2. Awareness of AI Act

This section explores the European DSOs' awareness of the AI Act and the support that the developers may require for its implementation.
The objective also includes evaluating whether DSOs have conducted risk-assessment analyses of their AI systems according to Annex III of the AI Act [5], due to their role in the management of critical digital infrastructure.
Despite AI's transformative potential to enhance operational efficiency and integrate renewable energy, many energy operators can lack in concrete strategies to comply with upcoming regulatory obligations [12]. Consequently, DSOs and AI developers can likely require technical and regulatory support.
The questions have been designed as follows:

1. Are you aware of the AI Act and its implications for AI applications in the energy sector?

2. Which challenges did you face in aligning your AI applications with the AI Act's requirements? (e.g. ethical, privacy, security, transparency, fairness, accountability issues…)

3. Would you need or do you foresee the necessity for any assistance to comprehend and apply the AI in your systems?

4. Has your organization taken steps to classify AI systems according to the risk levels defined in the AI Act?

5. Do you use high-risk AI systems for your grid?

As it can be seen from the questions, the scope here is to recognize the awareness level of the AI Act by the DSOs and identify challenges they may encounter. This way, possible guidelines for the AI Act usage can be circulated and more assistance can be planned to be given to the developers.

2.3. Use of the AI in the energy sector

AI is increasingly embedded into fields such as smart grid management, predictive maintenance, demand forecasting, real-time balancing, and renewable energy source (RES) integration. It involves data such as grid measurements (e.g. voltage, current, temperature), historical consumption data, weather and satellite data, and IoT sensor inputs. Pilot projects show that AI can make the electricity grid more reliable, better match supply with demand and use more renewable energy [13].
At the same time, an academic work [14] underlines the importance of maintaining human-in-the-loop supervision, decision traceability, and continuous model monitoring to mitigate model drift or data poisoning attacks.

This section aims to identify the areas where the DSOs use AI in the energy sector and if positive or negative effects have been identified. The following questions explores these aspects further.

6. Which areas do you currently use AI?
7. Did you expect any POSITIVE effects of AI in your energy operations?
8. Did you expect any NEGATIVE effects of AI in your energy operations?
9. Which algorithm does your AI system use?
10. Which data does your AI system use (e.g. historical data, other database – please specify type of data and time span that the data covers)?
11. Who judges the decisions/outcomes of AI systems? Please describe whether they can be effectively reversed or disregarded when necessary by the users.
12. Do you supervise the interaction of AI systems with other software and/ or hardware components?
13. How do you monitor for different types of misuse or unintended use of your AI systems, and what measures do you take to address them?

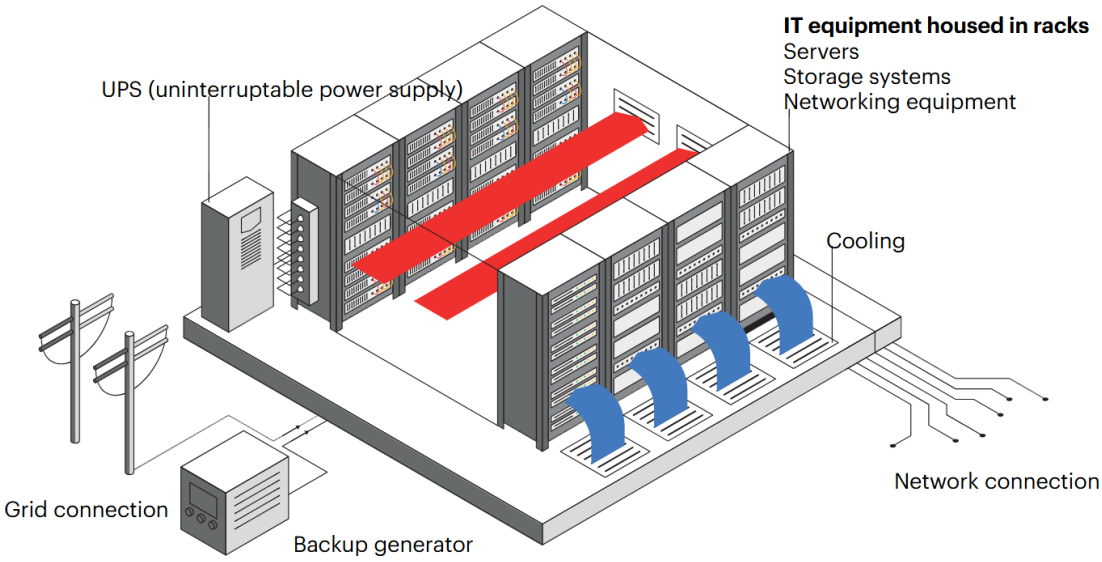
2.4. Energy efficiency and environmental impact

Assessing AI's environmental impact is challenging due to the significant electricity demands of data centres. In fact, these centres, which include servers, storage systems, networking equipment, and cooling systems (**Figure 1**), are experiencing rapidly increasing energy consumption.

According to the International Energy Agency report [11], data centres located in the United States, China and Europe together account for about 85% of global data centre electricity use, estimated at around 415 TWh in 2024, which represents roughly 1.5% of global electricity demand. Their energy consumption could more than double by 2030, mainly because of digitalisation and the increasing use of AI. AI-related operations already represent nearly one quarter of the electricity used by servers. In data centres, servers are responsible for most of the consumption (around 60%), while cooling and climate control systems can require up to 30%, depending on efficiency. This shows that, while AI can make energy systems smarter and more efficient, the infrastructure needed to

run it also consumes considerable amounts of energy, an aspect that DSOs and regulators are starting to include in their sustainability assessments.

Figure 1. Typical energy consumers in a data centre



Source: IEA, 2025 [11]

This section aims to address whether AI systems are used to mitigate the environmental impact of energy consumption, understanding whether the DSOs are taking actions in order to mitigate CO_2 footprint or actions to facilitate the uptake of RES, which would support environmental policies.

The questions of the questionnaire have been the following:

14. How does your organization use AI to optimize energy consumption and reduce environmental impact?
15. How do you assess the environmental impact of AI systems used in your operations (e.g. life cycle assessment, CO_2 footprint, ...?)
16. How do AI-driven decisions in your network contribute to RES integration?

2.5. Best practices, documentation and ethics

Under the AI Act, high-risk AI systems must ensure high-quality datasets, maintain traceability and guarantee effective human oversight. They must also maintain technical documentation demonstrating compliance, to be made available to competent authorities [15]. In addition, the report from the European Parliament Research Service [12] highlights that the ongoing digitalisation of the energy sector must be accompanied by robust governance and transparency.

Recent research [14] introduces frameworks like “E-TAI” (Energy-Trustworthy Artificial Intelligence), which adapts the general EU guidelines and emphasizes the explainability, auditability and accountability. In practice this includes: maintaining an AI system inventory (with risk levels and intended purposes), dataset documentation (sources, preprocessing, biases), clear role allocation for human operators (including override capacity), decision-path logging, transparent stakeholder

communication, continuous model monitoring, and energy-efficient model design. These best practices help build trust, compliance, and sustainability within AI-driven energy operations.

This section aims to assess the organization's technical documentation for transparency, regulatory compliance, and the promotion of sustainable AI practices. This includes instructions for the AI tools usage and data communication to interested stakeholders.

The questions that have been used are the following:

17. Do you document AI-driven decisions affecting energy distribution and transmission?
18. Do you include a detailed description of the AI systems, as required by the instructions for use?
19. How do you inform stakeholders and regulators about AI-based decisions affecting grid stability and consumer services?
20. How do you ensure that AI applications align with industry best practices?

2.6. Codes of Conduct

The AI Act also introduces voluntary codes of conduct for non-high-risk systems and promotes the establishment of regulatory sandboxes, allowing experimentation under controlled conditions [16]. Nonetheless, experts argue that the regulation does not yet fully address systemic risks arising from the interactions of multiple AI systems within interconnected energy grids [17].

Therefore, DSOs and technology providers are encouraged to develop internal codes of ethics covering transparency, fairness, environmental responsibility, and human supervision, even when not legally required. Before deployment, systems should undergo rigorous testing under realistic and stress-test scenarios, covering extreme events, data corruption, or cyberattacks, along with pilot phases, fallback mechanisms (human override, safe mode), behavioural monitoring, and comprehensive test documentation. Such pre-implementation testing and ethical codification not only mitigate technical and operational risks but also demonstrate proactive compliance and accountability, strengthening trust between regulators, operators, and end users.

This section evaluates whether the organization adheres to specific codes of conduct and performs pre-implementation testing. The questions have been the following:

21. Are you following any industry-specific codes of conduct for AI applications in energy?
22. Do you perform real-world testing of AI systems?

2.7. Social implications

AI adoption in the energy sector brings both opportunities and challenges for society. It can empower consumers through greater transparency, participation in energy communities, and behavioural change towards efficiency [16]. However, as highlighted by the European Parliament, increasing digitalisation may deepen inequalities in access to technology and raise data privacy

concerns. Moreover, the growing electricity demand of AI-related data centres could offset environmental gains. Ensuring inclusiveness, ethics, and public trust in AI-driven energy systems will therefore be essential for a socially sustainable transition [12].

This section assesses the social implications of AI systems. This would help policy makers understand the extent to which the DSOs apply AI tools addressing the social perspective.

23. Have you experienced, reported or noticed any social impact from the application of your AI systems?

2.8. Other remarks and future involvement

This subsection gives the opportunity to the participants to make further remarks on the survey and provide comments or suggestions for future research. It also addresses possible future involvement of the participants in the project. The respective questions are as follows:

24. Do you have any suggestions, remarks or feedback on this survey?

25. Are you interested in participating in future discussions, workshops, or initiatives related to AI systems in the energy sector? If so, please specify how you would like to be involved and provide your email address for follow-ups.

3. Survey analysis and results

3.1. Introduction

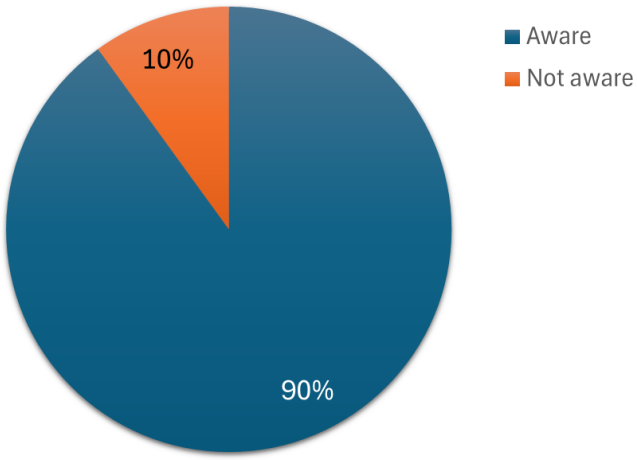
This chapter aims to offer insights on the AI Act awareness among DSOs, explore how AI contributes to energy efficiency, highlight best practices identified in AI applications, and illustrate the development of the code of conduct.

By examining these dimensions, the chapter provides a comprehensive overview of how DSOs are navigating regulatory landscapes while striving towards an intelligent and sustainable energy future. Policymakers can benefit from these insights to support the effective integration of AI across the sector, promoting innovation and compliance.

The JRC invited 30 DSOs to take part in the survey and received responses from 10. It should be noted that, given the limited number of responses (n=10), the results should be interpreted as indicative and do not aim to provide a statistically representative picture of all European DSOs.

Figure 2 shows that 90% (9 out of 10) of the responding DSOs are aware of the AI Act, with many beginning to integrate AI technologies to optimize operations and enhance customer service. Consequently, the way DSOs apply the AI Act provides a clear and practical example of how the regulation is being implemented in the energy sector. The only respondent not aware of the AI Act was a small regional DSO with limited deployment of AI technologies, which may explain the lower level of awareness.

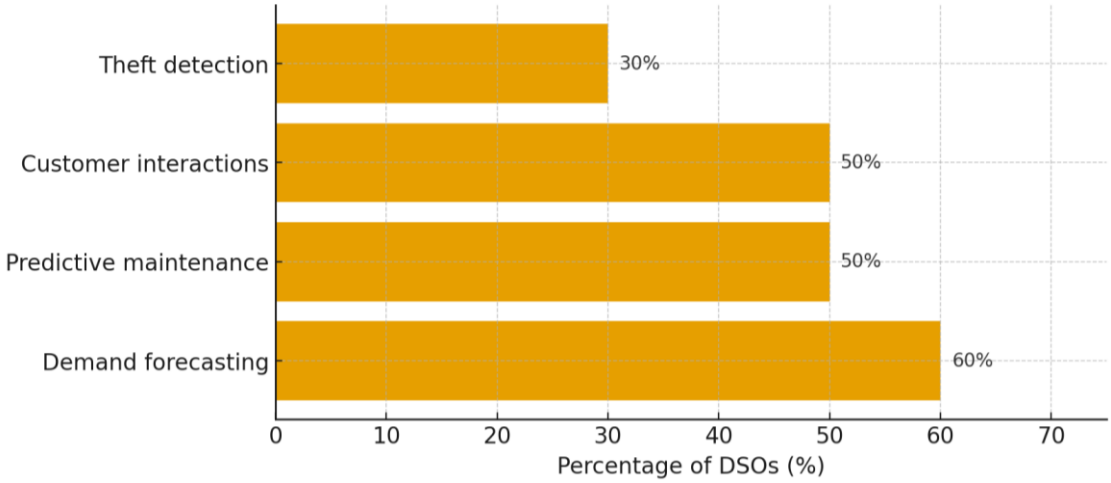
Figure 2. AI Act awareness among DSOs (survey n=10)



Source: JRC analysis

Figure 3 indicates that 60% (6 out of 10) of DSOs are using AI for demand forecasting, showing that companies are using data to take better decisions and work more efficiently. Furthermore, 50% (5 out of 10) are employing AI for predictive maintenance and customer interactions, highlighting that AI is also exploited to enhance the way services are provided. Additionally, innovative applications such as electricity theft detection are being explored by 30% (3 out 10) of DSOs, showcasing AI's potential to address diverse operational challenges.

Figure 3. Percentage of DSOs using AI applications in certain fields



Source: JRC analysis

3.2. Awareness and adaptation to the AI Act

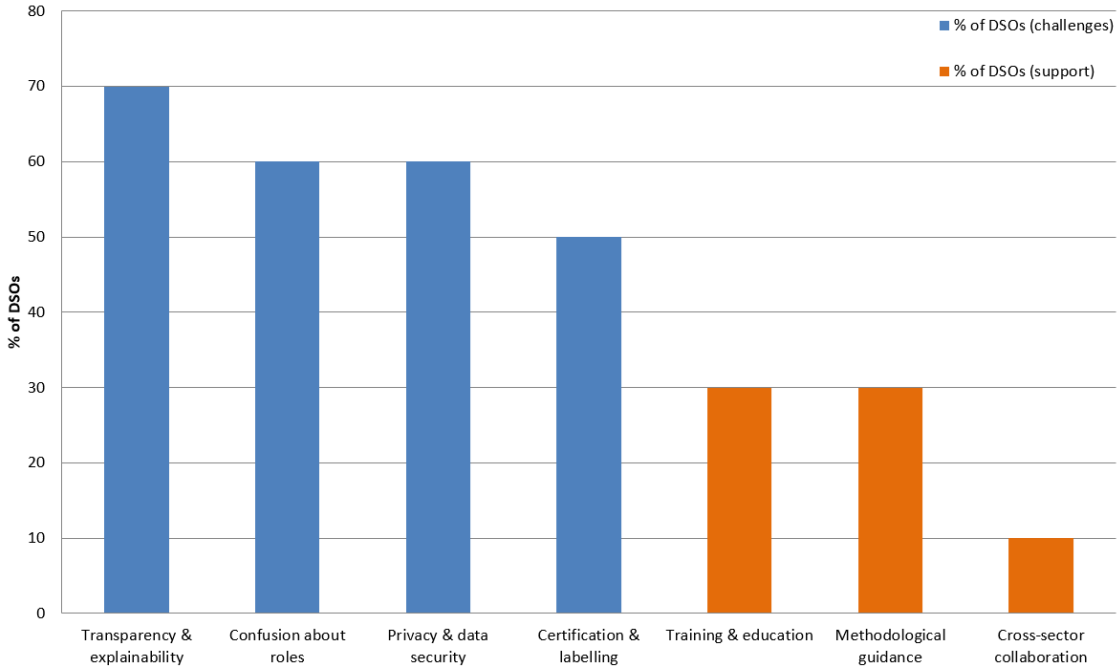
The awareness of the AI Act among DSOs is notably high, reflecting a significant understanding of the regulatory landscape governing AI implementation in the energy sector. The AI Act, designed to establish a legal framework that ensures AI technologies are safe and ethical, requires organizations to conform to standards that prioritize transparency, accountability, and risk management. As mentioned at the beginning of this chapter, 90% (9 out of 10) of DSOs said they know about these regulations, showing that the sector is committed to follow the rules and taking a responsible approach to digital transformation.

Although most respondents reported being aware of the AI Act, , many DSOs encounter challenges in interpreting some regulatory specifics. For example, terms like “deployer” and “provider” within the AI Act are causing confusion to 60% (6 out of 10) of DSOs analysed, as are the precise requirements for certification and labelling of AI systems. In addition to that, while awareness of the AI Act appears relatively high among the respondent DSOs, , the depth of understanding varies. Approximately 70% (7 out of 10) of respondents highlighted transparency and explainability as key challenges, particularly in relation to the “black box” nature of certain AI systems. Privacy and data security were also major concerns (60%) (6 out of 10), alongside uncertainties about regulatory definitions and certification procedures (50%, 5 out of 10). These findings point out that awareness is more conceptual rather than operational.

When asked about the kind of support needed to strengthen compliance, about 30% (3 out of 10) of DSOs emphasized the need for targeted training, educational initiatives and guidance on risk and impact assessment. Only one DSOs reported having an internal coordination framework, that involves several departments such as technology, legal, compliance, and audit to classify AI systems according to risk. Overall, these findings indicates that the sector would benefit from structured projects that can facilitate the adaptation to the AI Act.

Figure 4 presents all the challenges expressed by the respondents, and suggests that the foundational understanding is strong, but there is a need for more detailed guidance and educational initiatives. This would help DSOs move from awareness to practical compliance, ensuring that AI systems are used in line with regulations and good industry practices.

Figure 4. Challenges (blue) and support (orange) needs for AI Act compliance



Source: JRC analysis

3.3. Use of AI

Regarding the use of the AI among the DSOs, the survey shows a broad adoption of AI across multiple operational and strategic domains. **Figure 5** shows that demand forecasting and customer interactions are the most common AI applications, used by 60% (6 out of 10) of DSOs. Grid management follows at 50% (5 out of 10), while 40% (4 out of 10) also apply AI for predictive maintenance. In addition, DSOs reported exploring other AI applications, such as office support, knowledge management, recruitment, or specialized tasks like electricity theft detection.

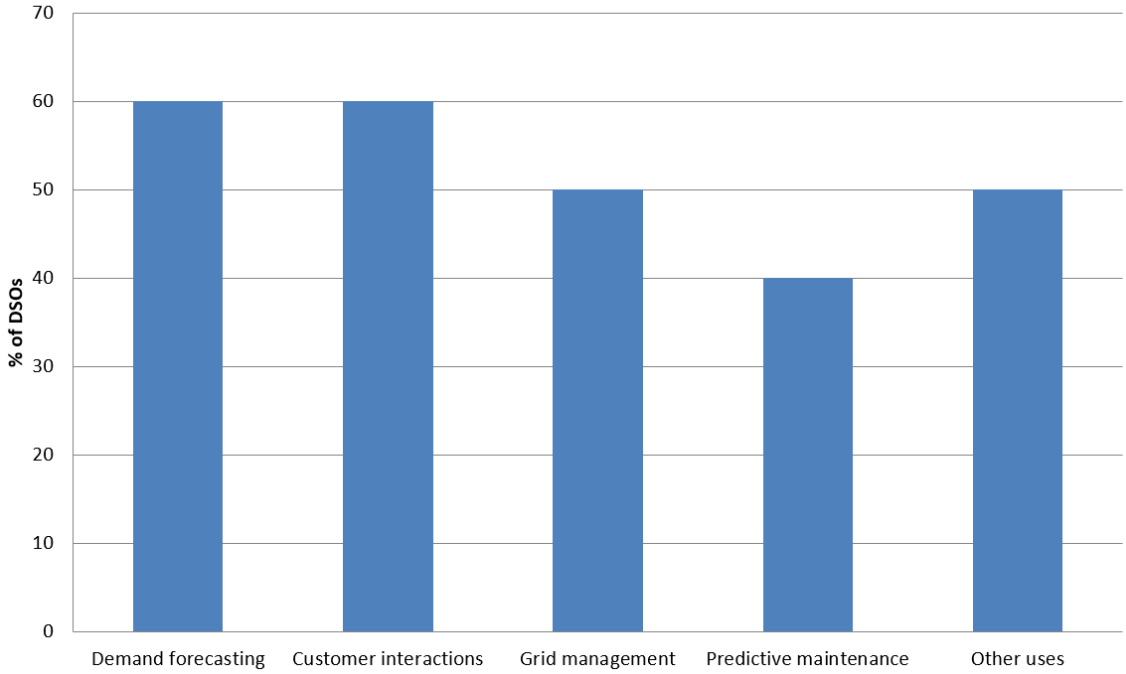
Positive impacts are acknowledged by most of the respondents 80% (8 out of 10) of DSOs report improved operational efficiency, faster and more objective decision-making, enhanced planning, and better customer engagement. Some DSOs highlighted ecological or sustainability gains, including reduced electricity losses and the successful implementation of ecological corridor management.

Despite these advantages, challenges remain even though they are less frequently expressed. In fact, 30% (3 out of 10) of DSOs mention issues such as increased data centre power consumption, security concerns, and human-AI interaction adaptation to AI-driven changes, including fears of reduced workload or lower quality of decisions.

The algorithms deployed among the DSOs vary according to the business case, including neural networks, decision trees, clustering methods, time-series analysis, deep learning, large language models, and expert systems. Data inputs are diverse, encompassing smart meter readings, historical operational data, GIS data, sensor data, and internal documentation.

Also, AI deployment is accompanied by consistent human oversight in almost all cases: outputs are reviewed by experts, human-in-the-loop systems, or dedicated teams with the authority to adjust, override, or disregard AI recommendations.

Figure 5. Distribution of AI use cases among DSOs



Source: JRC analysis

Governance, supervision, and misuse monitoring are implemented by all the DSOs through centralized AI management, IT logging, cybersecurity measures, external audits, preventive controls, and staff training programs.
Overall, AI is primarily used to support operational efficiency, strategic planning, and customer service, while all DSOs explicitly avoid relying on AI for the direct operation of critical infrastructure, emphasizing its role as a decision-support tool rather than a replacement for human judgment.

3.4. Energy efficiency and environmental impact

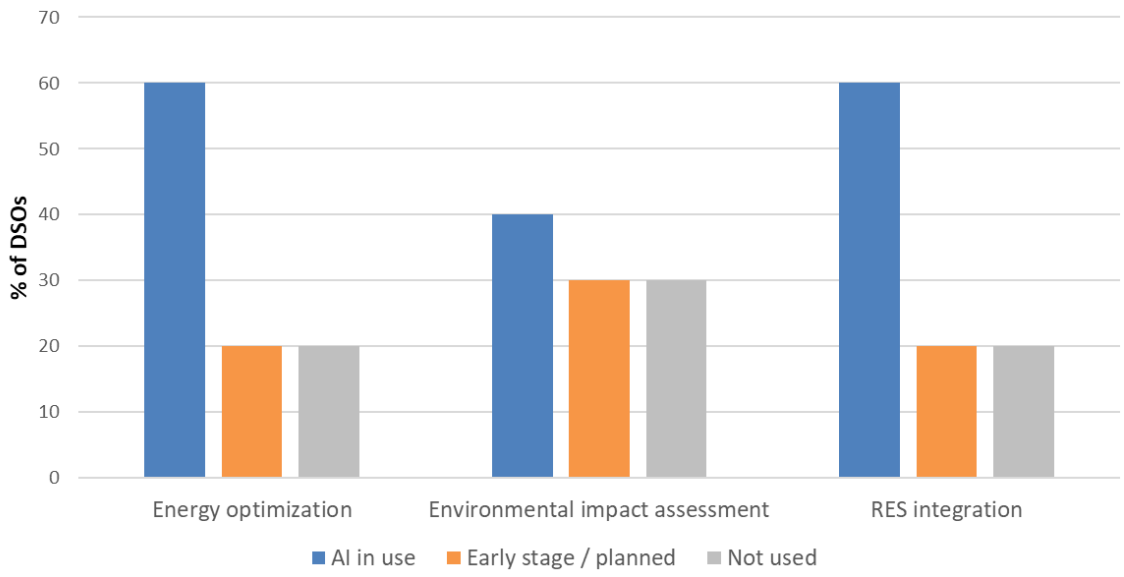
The AI adoption among DSOs shows increasing attention to energy optimization, environmental monitoring, and renewable integration (see **Figure 6**). 70% (7 out of 10) of respondents report at least some form of AI use related to energy efficiency or sustainability, although the level of maturity varies significantly across organizations. The most common applications concern demand forecasting, green IT practices, and grid optimization. Several DSOs use AI for short-term electricity consumption forecasting, fault detection, and anomaly identification through drone-based inspections or deep learning models. One DSO applies AI for electricity theft detection, directly reducing technical losses, while another uses smart assistants for real-time energy monitoring and route optimization. In other cases, AI contributes indirectly to sustainability through efficient IT infrastructure, reduced manual workloads, and lower resource consumption.

Regarding the environmental impact assessment, half of the DSOs have started to integrate the sustainability monitoring into their AI systems. Some organizations include AI-related metric (such as energy consumption of AI models, CO_2 emissions from computing, energy efficiency gains) in their CSRD reports (Corporate Sustainability Reporting Directive). In some cases DSOs directly monitor the energy use of their AI models, particularly those deployed on cloud infrastructures

powered by renewable energy through Power Purchase Agreements (PPAs) which are long-term contract between a (usually) renewable energy producer and consumer (e.g. DSO or a company). These contracts guarantee that the energy used for the AI services is green, help reduce the carbon footprint and support the company's sustainability goals. However, for several operators, environmental impact assessment remains at an early or preparatory stage, mainly due to the limited scale or experimental nature of current AI projects.

With regards to the renewable energy systems integration, 60% (6 out of 10) DSOs report ongoing or planned activities. AI is being used to support the integration of prosumers and renewable generation by enhancing grid flexibility, improving capacity planning, and optimizing the placement of energy storage systems. Some DSOs employ AI to profile renewable energy sources in the grid or to predict photovoltaic production and flexibility needs. One of the DSO describes an advanced initiative, which uses AI to improve how prosumer (i.e. customers that produce and consume electricity) are managed. The project aims to make the connections requests, monitoring energy flows and managing metering and billing processes more efficient, ensuring a smooth generation into the grid. The expected benefits include better compliance with renewable energy regulation, faster and more transparent handling of prosumer requests, and improved customer satisfaction. Others, however, underline that AI solutions in this domain are still immature and that expert judgment remains essential for reliable forecasting and decision-making.

Figure 6. AI adoption in energy efficiency and sustainability



Source: JRC analysis

Overall, the DSOs acknowledge AI as a key enabler of energy transition, capable of optimizing grid operations and supporting sustainability objectives. However, the environmental impact monitoring and renewable integration through AI are still in an evolving phase, with most applications focused on energy efficiency and predictive capabilities rather than directly reducing greenhouse gas emission.

3.5. Best practices

The maturity level in documenting, monitoring, and communicating AI-related practices varies significantly among different Distribution System Operators (DSOs). Some operators have established comprehensive procedures for documenting AI decisions, ensuring traceability and transparency in the development and deployment of AI systems. This includes maintaining detailed records of decision-making processes, data sources, algorithms used, and the rationale behind AI-driven decisions. Such detailed documentation facilitates and enhances both accountability and trust among stakeholders, providing clear insights into AI operations.

In contrast, other DSOs are still in the early stages, where documentation efforts are either undeveloped or confined to limited pilot projects. This lack of comprehensive documentation can hinder the understanding of AI impacts and may undermine efforts to build trust and accountability. It underscores a gap in the adoption of standardized practices that ensure AI operations are aligned with industry norms and regulatory expectations.

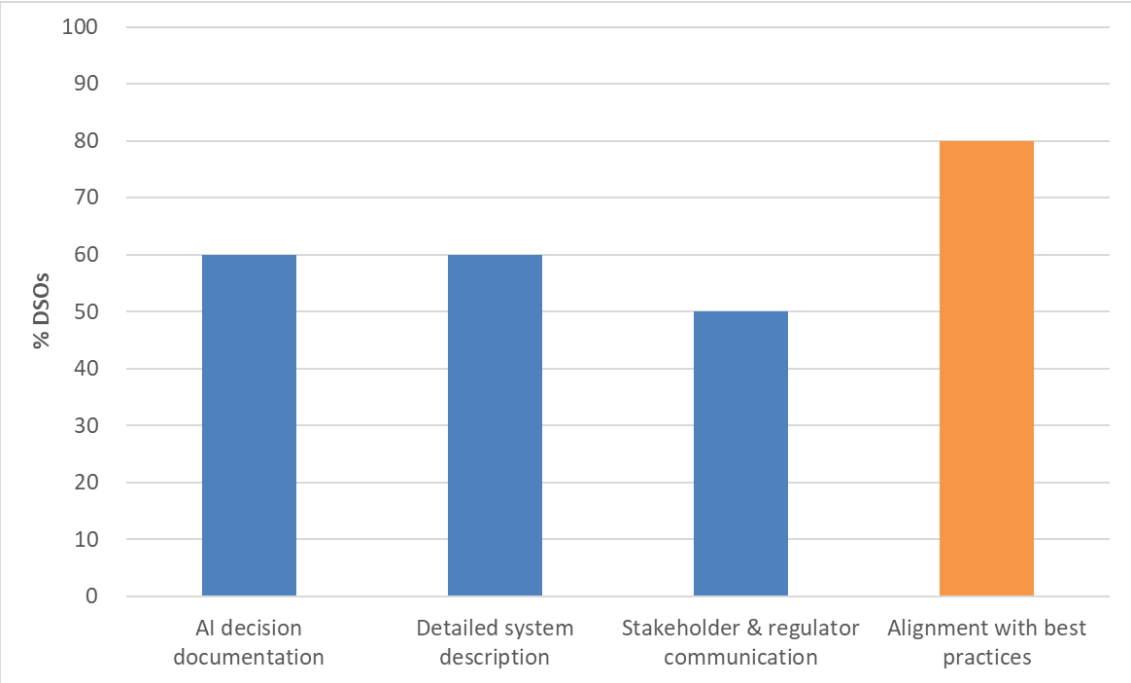
The presence of a detailed AI system description often indicates organizational maturity. DSOs that have centralized systems to control AI systems, like dedicated teams or internal guidelines, tend to manage their AI activities more consistently and securely. This includes having important positions like Data Protection Officers (DPOs) and Chief Information Security Officers (CISOs) who ensure that all records and data related to AI are properly maintained and protected. These governance frameworks facilitate a cohesive approach across the organization, ensuring that AI initiatives are not only well-documented but also align with the organizational objectives and regulatory requirements. These structures provide a solid foundation for a culture of accountability and transparency, ensuring that AI is integrated responsibly across the organization.

For what concern the communication with stakeholders and regulators, only few DSOs have established systematic procedures. These may include routine public disclosure of load forecasts or the publication of regular sustainability reports, providing stakeholders with a clear and consistent view of the DSOs' AI activities and their impacts. Most DSOs, however, limit their engagement to privacy statements or project-specific reports. This communication approach might not fully convey the scope and implications of AI initiatives, potentially leading to misunderstandings about their intentions and outcomes.

Finally, alignment with best practices demonstrates a strategic and forward-looking approach. Several DSOs actively participate in cross-industry collaborations, internal networks, and partnerships with academia or technology providers. These collaborations foster innovation and ensure consistency with evolving regulatory and ethical standards. Participation in such networks allows DSOs to share knowledge, explore new AI technologies, and establish a leadership position in the ethical and responsible adoption of AI. By actively engaging in these initiatives, DSOs not only drive technological advancements but also contribute to setting industry standards and best practices, reinforcing their commitment to responsible AI development.

Figure 7 below illustrates the relative adoption of these practices across DSOs, showing where documentation, communication, and alignment with best practices are more advanced. Alignment with best practices appears the most consistently applied dimension among surveyed DSOs. The alignment dimension is highlighted in orange to emphasize its proactive nature, as it represents the link between compliance, innovation and improvement.

Figure 7. Percentage of DSOs with positive practices



Source: JRC analysis

3.6. Codes of conduct

Most DSOs have not adopted a formal AI Code of Conduct or similar internal guidelines for deploying AI. Only few of them, have internal standards addressing AI development, security, and employee responsibilities. Notably, one DSO has collaborated with external consultants and public stakeholders to establish comprehensive company standards for AI strategy and governance.

Furthermore, some DSOs ensure they meet ethical and regulatory standards by routinely consulting internal security and data protection officers, even without a formal code. This shows a practical approach to making sure everything is in order before deploying AI.

Regarding real-world testing, many DSOs already engage in pilot testing or field trials before fully rolling out AI systems. For instance, some of them are conducting on-site evaluations of AI solutions with technicians, while others test the system on a small portion of the network, such as a limited number of devices or a specific geographic area, to ensure each AI application is reliable before official deployment. One of the DSOs respondents is cross-checking AI forecasts against traditional methods on daily basis, stopping AI outputs when significant discrepancies are detected

In summary, while formal AI Codes of Conduct are not common, many DSOs use alternative governance methods, like pilot testing and internal reviews, to ensure AI is used responsibly and securely.

4. Conclusions

The report aimed to explore the current state of awareness, application, and governance of the AI Act among selected European DSOs, as well as its implications. It emerges that the AI Act is becoming a key framework, laying the foundation for safer and ethically compliant AI technologies within the energy sector. Despite some DSOs being aware of the AI Act and beginning to integrate AI technologies to enhance operational efficiency and customer service, several areas still require improvement, such as the formal implementation of AI codes of conduct and clearer interpretation of regulatory provisions.

Even though there are no explicit AI codes of conduct, DSOs are employing a variety of governance mechanisms like pilot testing, internal reviews, and compliance checks to ensure responsible AI use. These approaches show a pragmatic commitment to achieve both ethical and regulatory standards, even without formal guidelines.

The survey results reveal that DSOs are increasingly applying AI technologies across various domains, such as demand forecasting, predictive maintenance, and customer interactions, enhancing energy efficiency and supporting sustainability goals. However, the environmental impact of AI is still rarely monitored, suggesting that further attention is needed in this area. In addition, social challenges such as data privacy, regulatory interpretation, and human-AI interaction remain important areas requiring careful consideration.

Policymakers can leverage these insights to reinforce AI's integration across the energy sector, encouraging innovation, while ensuring compliance with evolving regulations. The active participation of DSOs in collaborations with industries and academic partnerships reflects their commitment to maintaining a competitive edge in the ethical adoption of AI.

To address identified challenges, targeted training, methodological guidance, and structured capacity-building programs are crucial. These initiatives will support the transition from theoretical understanding to practical and effective integration of AI, driving substantial advancements in the energy sector.

In conclusion, fostering an environment where best practices are shared and adopted will empower DSOs to lead in responsible AI usage. As the energy sector evolves, embracing AI as a catalyst for intelligent, sustainable growth will be vital to support the energy transition and achieve climate and sustainability goals.

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List of abbreviations and definitions

Abbreviations	Definitions
AI	Artificial Intelligence
CISOs	Chief Information Security Officers
CO_2	Carbon Dioxide
CSRD	Corporate Sustainability Reporting Directive
DMA	Digital Markets Act
DPO	Data Protection Officer
DSA	Digital Services Act
DSO	Distribution System Operator
EU	European Union
GDPR	General Data Protection Regulation
GIS	Geographic Information System
JRC	Joint Research Centre
RES	Renewable Energy Systems
PPAs	Power Purchase Agreements
USD	United States Dollar

List of figures

Figure 1. Typical energy consumers in a data centre 13

Figure 2. AI Act awareness among DSOs (survey n=10) 16

Figure 3. Percentage of DSOs using AI applications in certain fields 17

Figure 4. Challenges (blue) and support (orange) needs for AI Act compliance 18

Figure 5. Distribution of AI use cases among DSOs 19

Figure 6. AI adoption in energy efficiency and sustainability 20

Figure 7. Percentage of DSOs with positive practices 22

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